

perform thorough checks, such as design rule checks (DRC) and layout versus schematic (LVS) verification.

9. Tape-out and tape-in. Tape-out and tape-in processes involve preparing the final design for fabrication and receiving the fabricated chips. Engineers specialising in these processes manage the transfer of design files to the foundry, collaborate with foundry partners, and ensure successful chip production. “We give certificates only if they have done tape-out. Writing on a resume that you have done a tape-out, I think companies look for these kinds of people who have done silicon tape-outs,” Kunal says.

Understanding the responsibilities and expertise required in each job role is crucial for individuals seeking a career in electronic design and for organisations aiming to develop innovative and efficient electronic products. “Every student goes through all these training sessions to get a VSD certificate. This is basically to bring out what their interest is. So, they have to go through all of it, and then they will identify what they’re interested in. For that field, they can either go for higher studies or some other institute that takes care of that particular field with the commercial tools,” he says.

VLSI, embedded systems, firmware: expanding opportunities

The field of VLSI is undergoing a transformation as it expands to encompass embedded systems and firmware, raising questions about the job opportunities and roles that arise from integrating these elements. Let’s delve into the relationship between VLSI, embedded systems, and firmware, exploring how these domains converge and the implications for job prospects.

Expanding beyond VLSI. Traditionally, VLSI engineers focused

The open source foundry kit and development of SoC

The emergence of open source foundry kits has brought about a transformative shift in the chip design world, creating new opportunities for learning and innovation.

- These kits consist of essential files and libraries, opening up new opportunities for learning and innovation in chip design.
- The integration of the SkyWater open source foundry kit has facilitated chip design education, despite the proprietary nature of chip design libraries.
- Collaborative partnerships between SkyWater, Google, Efabless, The OpenROAD Project, and the University of California, San Diego, have formed the foundation of the open source foundry ecosystem.
- RISC-V processors play a crucial role within this ecosystem, offering flexibility and adaptability to chip designs.
- The open source foundry ecosystem promotes accessibility by requiring minimal infrastructure and providing pre-installed image files for easy access to necessary tools.
- This accessibility encourages the democratisation of chip design and fosters broader community participation in the innovation process.

The open source foundry ecosystem is revolutionising the chip design field, empowering individuals to explore and innovate while promoting inclusivity. “After successfully constructing the SoC, the designer possesses their own custom chip,” Kunal says.

on chip design and fabrication, and completing chip manufacturing signified the conclusion of their primary responsibilities. However, a critical question emerged: What comes next? Embedded systems emerged as a potential answer to this query. The interface between the board and the chip is at the core of this transition. This interplay offers significant advantages, enabling a comprehensive understanding of the chip’s internals, such as the RTL code. Moreover, the ability to write software to initiate and operate the chip marks the domain of firmware design engineers.

Unlocking the potential of embedded systems. Embedded systems play a pivotal role in realising the full potential of chips. The hardware board serves as a bridge between the chip and its practical applications. With this board, engineers gain visibility into the chip’s internal components, allowing for a deeper exploration of its functionalities. The board’s capacity to reveal the RTL code of the chip facilitates thorough analysis and customisation. Leveraging this insight, engineers can develop software to boot up

the chip, harnessing its capabilities. The integration of embedded systems augments the role of VLSI engineers, expanding their expertise beyond chip design and delving into firmware development.

Diverse job prospects. The fusion of VLSI-embedded systems and firmware widens the horizon for job prospects in the field. While VLSI expertise forms the foundation, additional proficiency in embedded systems and firmware paves the way for diverse opportunities. Companies seek professionals with a comprehensive skill set capable of seamlessly navigating these domains. The integration of hardware and software aspects and the ability to customise chip functionalities make individuals well-equipped for various positions in chip design, system integration, firmware development, and application-specific roles.

The convergence of VLSI, embedded systems, and firmware brings an exciting era of diverse job prospects. Moving beyond chip design, engineers explore the potential of embedded systems, where the interplay between the board and chip unlocks new avenues for innova-